Korobochka (コロボ) 🇦🇺✝️🇷🇺 @cirnosad

Mar 4, 2024 • 31 tweets • cirnosad/status/1764649127231820278

Psychophysics

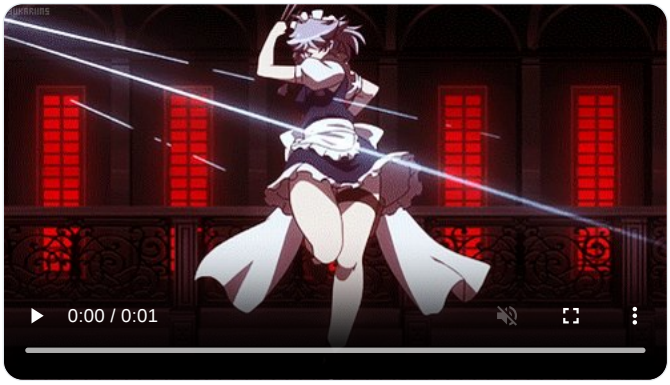
Most people see explosions and they don't appreciate the scale and type. There's no sense of scale based on pri/secondary effects!

This thread is all about the power law and our logarithmic senses, affecting sight, hearing, time and event perception of events!
/ 🧵

Let's start with explosions! In a previous thread I asserted that when a non-shaped charge explodes, ignoring things like reflections from the ground, buoyancy and gravity, the explosion will tend to go in all directions! Why is that, logically?

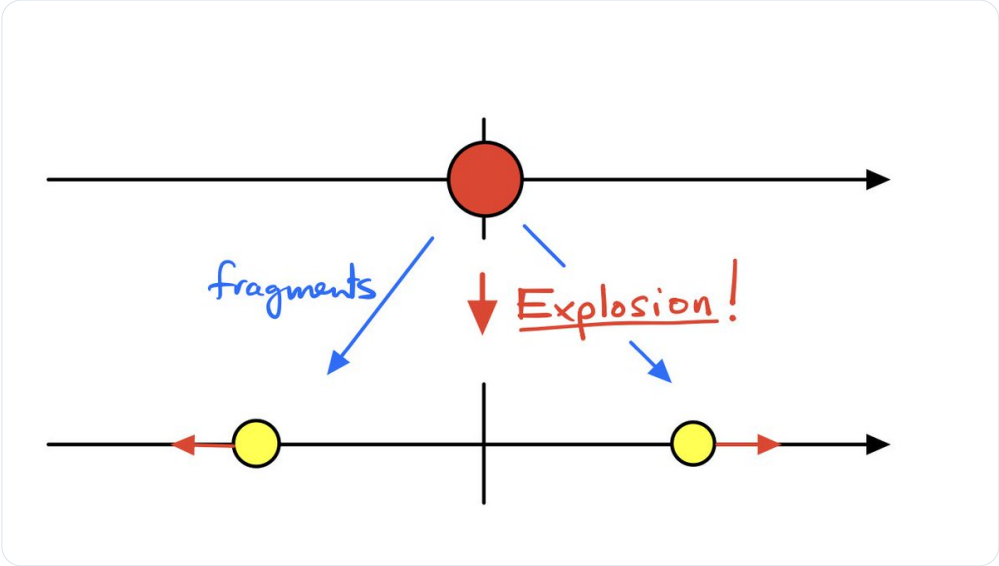
https://twitter.com/cirnosad/status/1659806116187099136

Simply put, there would be no preference of direction!
The explosion almost always arrises out of random collisions, which have no directionality.
If something is to blow up in all directions, it would do so with approximately the same intensity everywhere!
Let's start out in 1D!



If we were somehow in a 1D (one dimensional) universe and had a 1D explosive, it would have to conserve momentum and would produce two identical projectiles travelling in opposite directions or a chunk that is bigger than the other, moving slower.

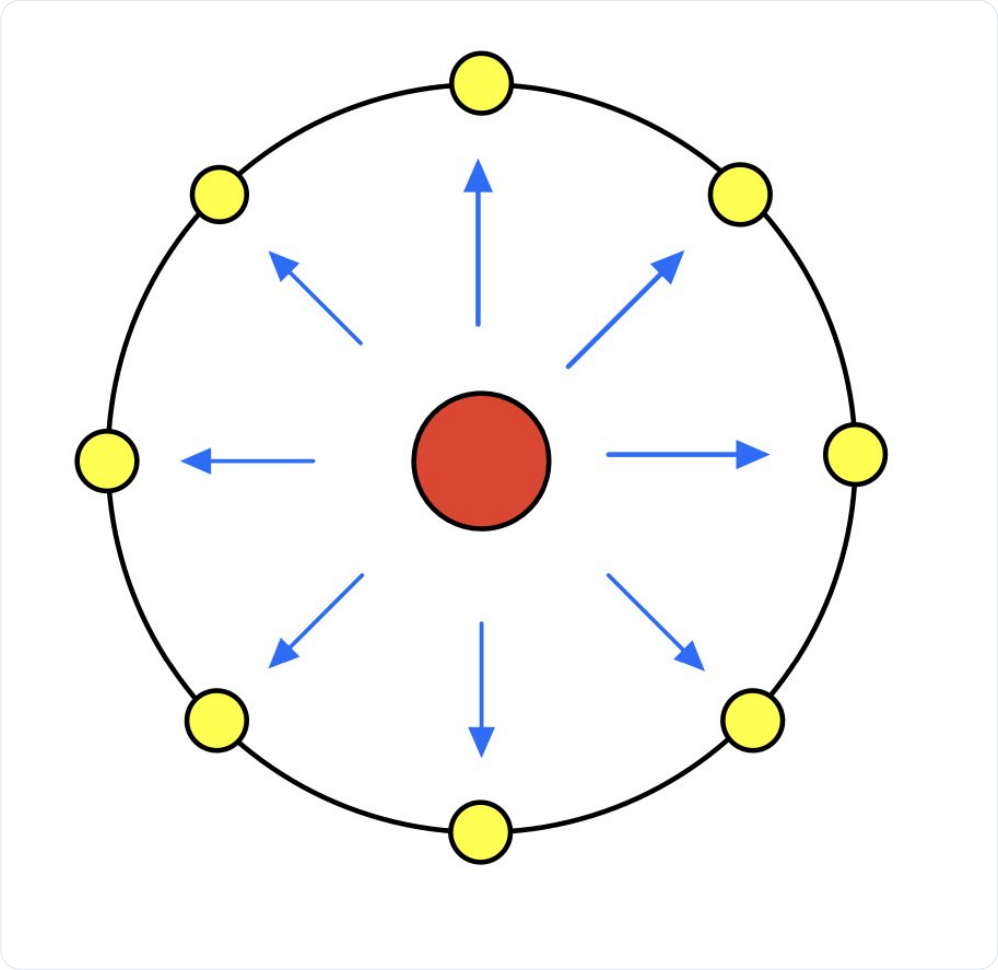
But on average it's the same!



Those two pieces would be travelling forever without losing any speed in empty space.

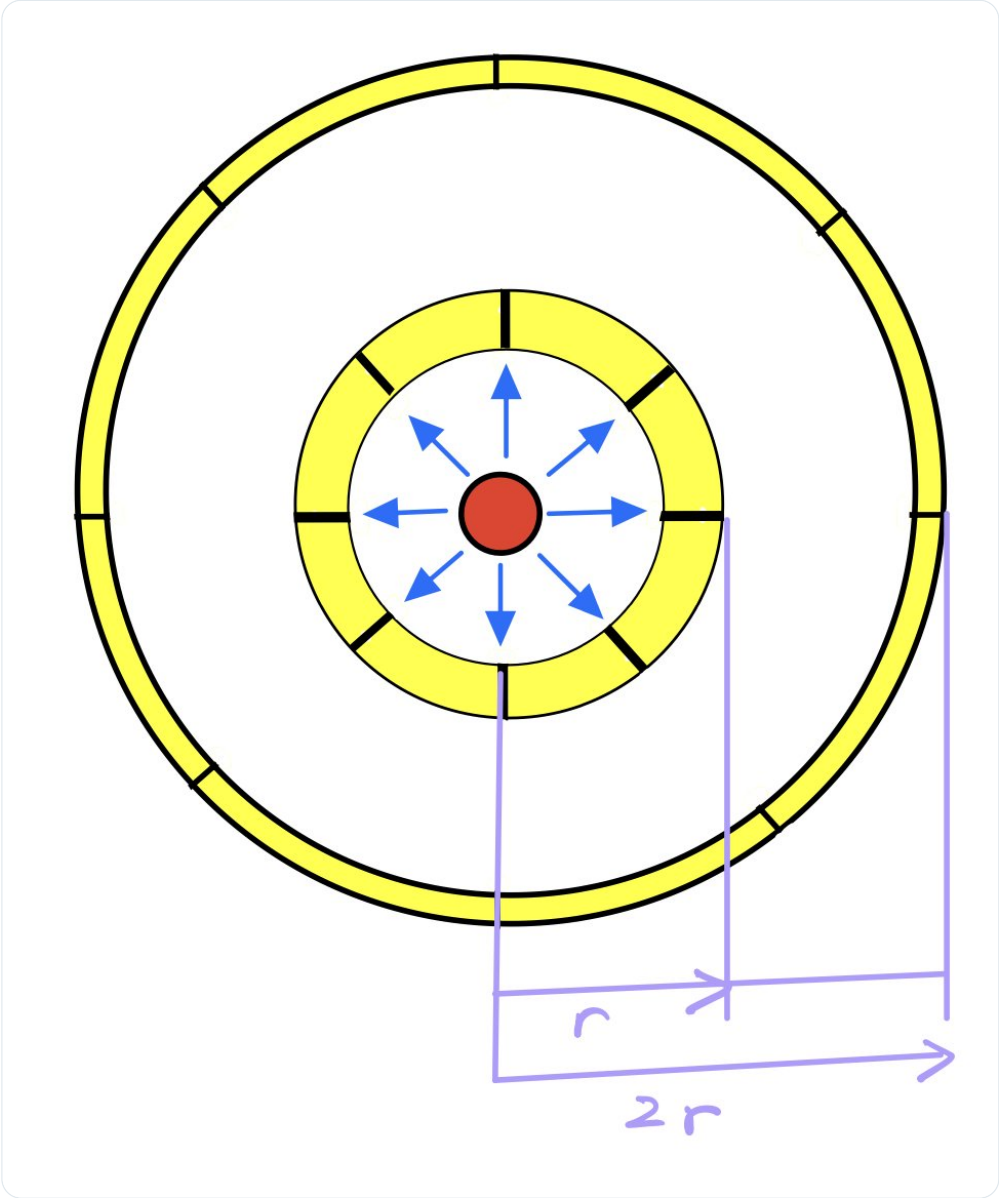
What if we lived in a 2D universe? We carry this idea onwards: on average every piece produced by the explosive is spread out in opposite directions.

It generates this shockwave circle:



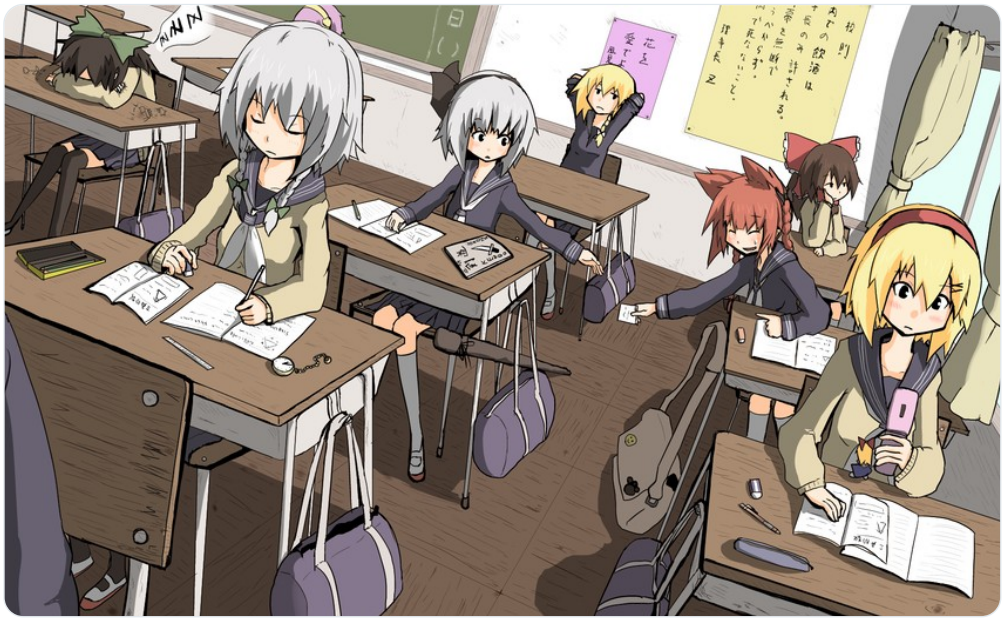
But of course in 2D it wouldn't be that simple. The explosive reaction wouldn't just create chunks, it would create a fine material/gas that looks much more like a wave.

As the wave move onwards in circles, it gets thinner as it expands outwards. Here we have two distances, r, 2r



Consider each arc section of this poorly drawn diagram. This is meant to contain the same amount of debris (it's the same thing, moving forward). As it moves from distance r to a distance 2r, the debris will become half as "thick". Proving this is simple, from primary school!

What's the circumference of a circle?
Two times Pi times r. (2 pi r).
If the material around the inner ring initially has thickness d, the area of the material would be approximately Area = 2 pi r d. If the material has constant density, the area shouldn't change as it moves.



At a distance 2 r, that means we must still have the same area. What's the next thickness d'? Let's do some balancing.
Area = 2 pi r d = 2 pi * 2 r d'

Eliminating the common factors from both sides:

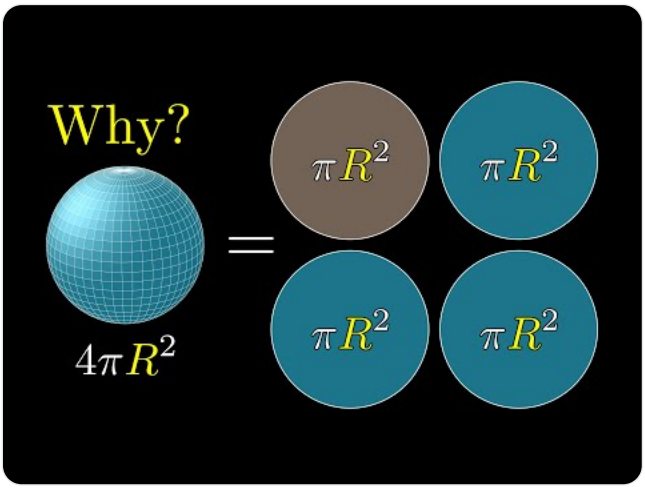
d' = d/2.

Thus, twice the distance means half the density in 2 dimensions!

What about in 3 dimensions? It's actually the exact same idea. We just need a formula for the surface of a sphere.

Why a sphere? Just as circles represents no directional preference in 2D, spheres represents no preference in 3D.

The equation is 4 pi r^2:



https://www.youtube.com/embed/GNcFjFmqEc8

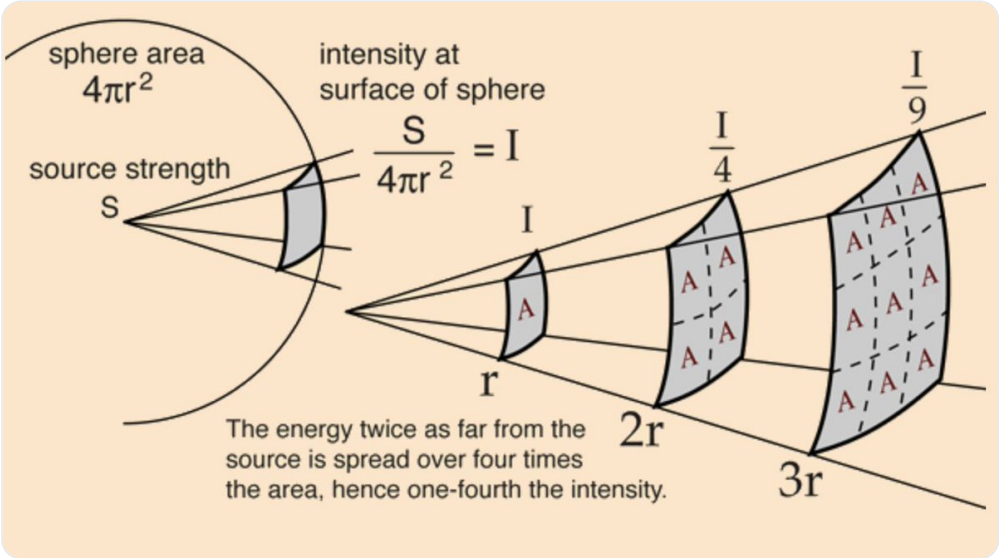
Thus, considering two distances again, r and 2r:

Area = 4 pi r^2 d = 4 pi (2r)^2 d'

Eliminating common factors:

d'=d/4.

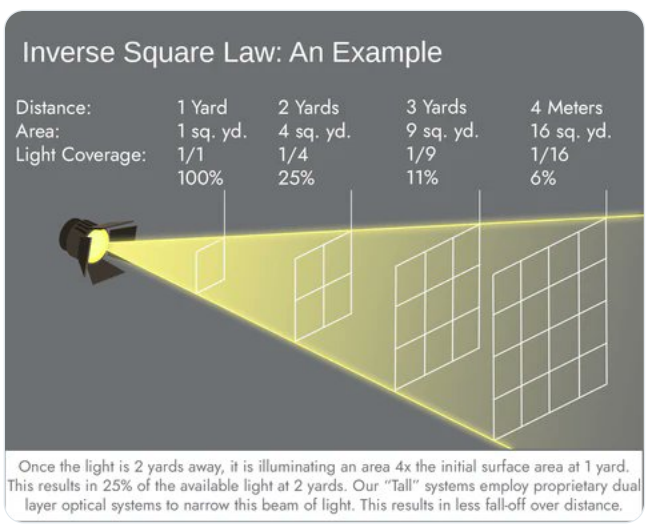
Thus in three dimensions, going twice the distance means losing four times the thickness of the shockwave!



In general, this forms a power-law, the most common being the inverse-square law (1/r^2 law). This isn't limited to just this explosion we discussed but almost EVERYTHING you experience every day:

- Light and energy
- Sound
- Gravity
- Electricity
- Contamination

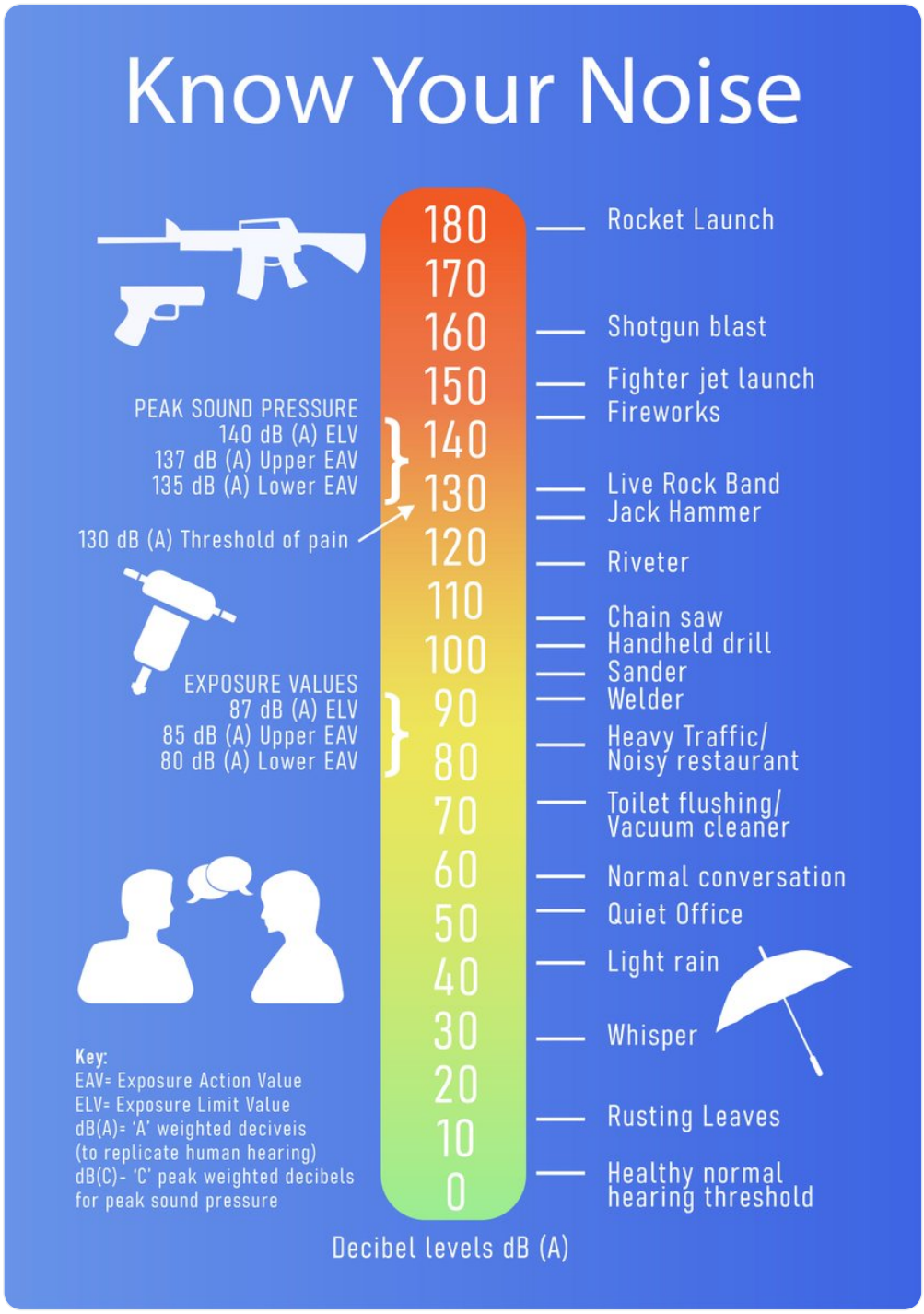
And more!



This may seem counter-intuitive to you! When you get twice as far from a light bulb, it doesn't get four times dimmer. I mean of course it gets dimmer overall, but no where near that much. Then how?
You see,we humans are designed with a powerful principle to deal with power laws.

Enter Weber-Fechner laws. What researchers realised is, as human beings we can only compare the relative difference between measurement, but in magnitude. Logarithmically.

You probably have heard of this before. Remember the dB scale? Let's refresh our memory on this one.



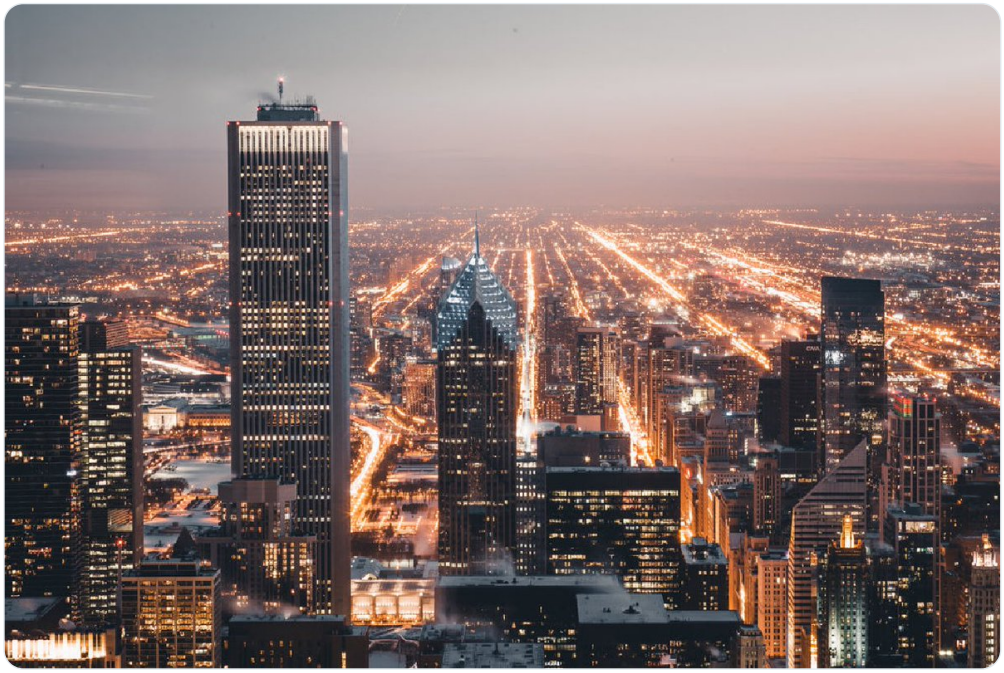
A welder is only +10dB louder than a flushing toilet. But is it in fact 10 times more powerful. In comparison to a quiet office, it's +40dB louder. But this means it's 10,000 more powerful!

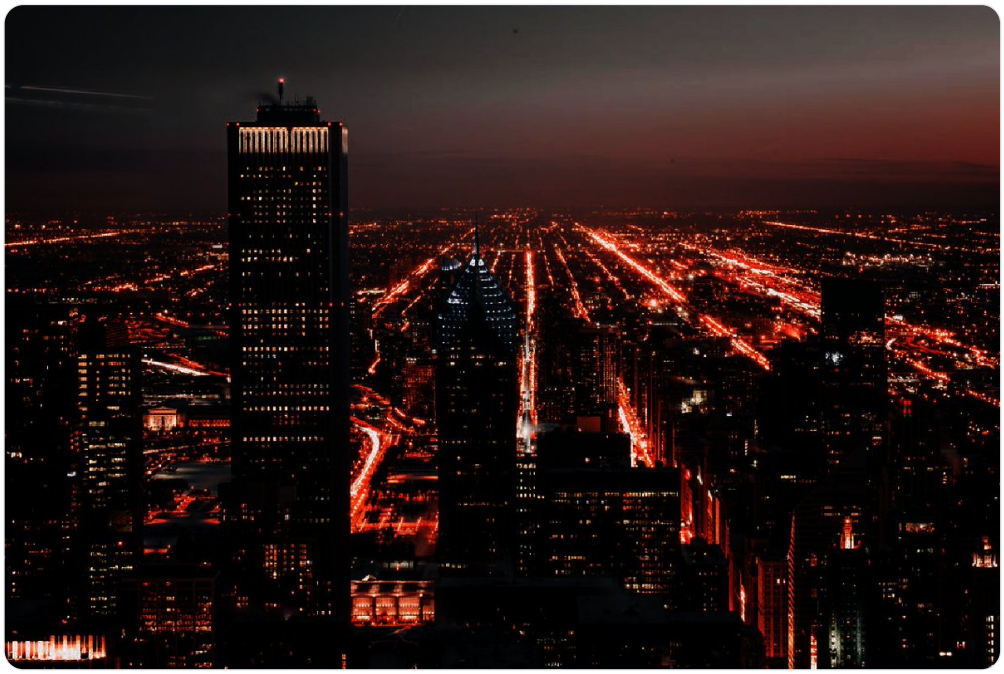
This is because this is a log-scale, which matches how our senses work!

To put the concepts together: This is why you can seriously hurt your ear drums with inner ear buds. By creating a pressure wave so close to your ear, 5 millimeters away, compared to hearing it 5 meters away, you generate a sound 1,000,000 more powerful (+60dB) in your ears!

Even though our perception is logarithmic, the world around us works on the actual value of the intensity of the sound wave, the density of the explosions, the intensity of the light and so on.

This isn't all bad, however. Look how a city would look like if we saw intensity:





But it's still something we must be aware of, we only perceive things on a relative scale to observations near them.

This is not limited to physical phenomena.

It's also how we perceive time. We compare it to our prior experiences, and time seems to "go faster" as we age.

focus

Why do we perceive logarithmically?

Why do small children place 3 halfway between 1 and 10? Why do two light bulbs not seem twice as bright as one? Why do we perceive so many things logarithmically? **Lav R. Varshney** and **John Z. Sun** can explain it, through evolution, the statistics of nature – and reducing error.

[By the way, time doesn't have to pass faster as you age, you can make time slow down, you just need to change your perception and do new things very frequently. You can make a day last a lifetime!

Sometimes, when something immensely significant happens, it seems time stops.]

Now let's get back to explosions.
Let's look at this grenade (Mk2) and this JDAM 2000 pound bomb (Mk84).
One contains 52 grams of filling. The other contains 428 kilograms of filling.
Getting near a JDAM or a grenade is a bad idea, but how close do you have to be to die?

This is referred to as the lethality radius.
Note beforehand, the JDAM is 10,000 times more heavy than the grenade.

The lethality radius of a Mk84 is 360 meters.
The lethality radius of a Mk2 is ~7 meters.

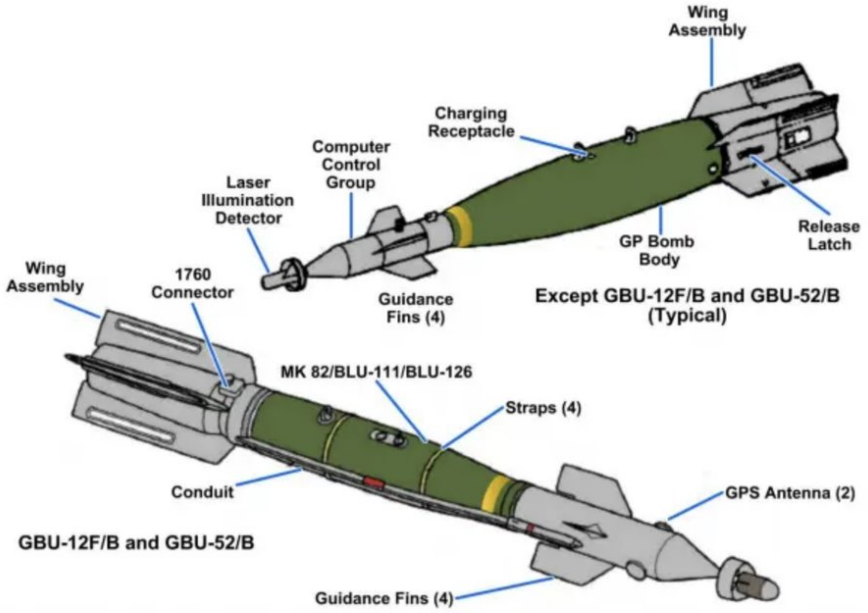
Despite being 10,000 heavier, it only has 50x the lethality range.

The fragmentation pattern and range of a 2,000lb Mk 84 bomb are difficult to predict, but it is generally said that this weapon has a 'lethal radius' (i.e. the distance in which it is likely to kill people in the vicinity) of up to 360m.

The blast waves of such a weapon can create a very great concussive effect; a 2,000lb bomb can be expected to cause severe injury and damage as far as 800 metres from the point of impact.

The wide-area impact of these weapons can be affected by a range of tactical, technical, and environmental factors, including fusing and the angle of attack.

Physical Characteristics	GBU-12/GBU-51/GBU-52	GBU-16	GBU-10
Weight	610 lb.	1100 lb.	2081 lb.
Dimensions:			
Length	131 in.	145 in.	170 in.
Diameter	11 in.	14 in.	18 in.
Suspension Provisions	14 in.	14 in.	30 in.



More characteristics of the GBU series.

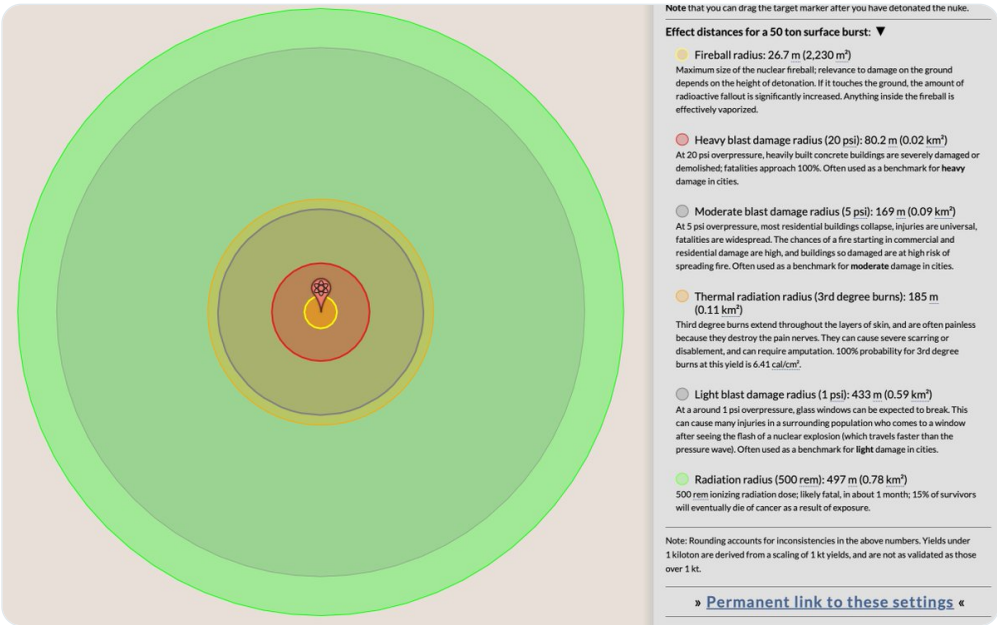
Why is that? Modelling the lethality range in a straight-forward way isn't simple, because they have two modes of lethality, but the r^2 law gets us to 2500, which is close to the ratio of weights!

Basically, to increase the lethality radius by 2, yield must be multiplied by 4.

Believe it or not, this extends to nuclear bombs (you knew I was going there). To increase the lethality range to 3.6km, you would need roughly 100 times the yield of a Mk84, a 50 ton lower yield nuclear weapon. But it's quite optimistic since nukes don't work on fragmentation...

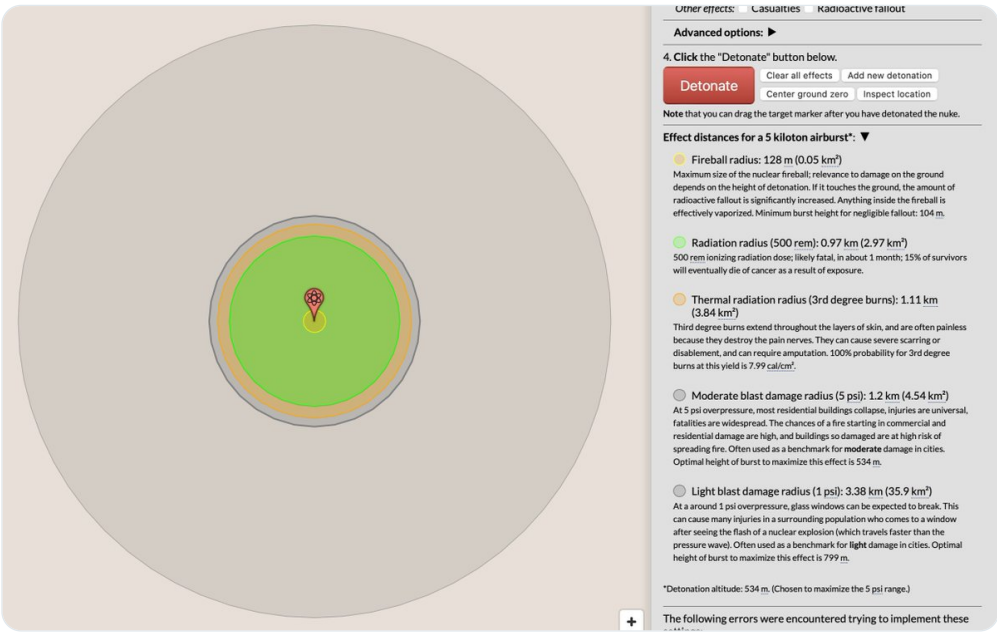
You would in fact, need a lot more! Most of the energy of the nuclear bomb would be directed up into the sky, and the lethality range is likely in the 400 meter range for a surface detonation and in the 500 meter range for an airburst.

What if we go 100 times bigger?



Here's what a 5kT air burst would do... the lethality range becomes around 3km. Devastating but probably not as bad as you thought.

On Earth, nukes actually scales a bit worse than r^2 due to absorption effects and other complex non-linearities.



You see, not everything about a nuclear explosion involves an outwards wave.
The fireball itself forms a very hot plasma, which stays together and is lifted up by buoyancy. This is related to the volume of the fireball, implying an r^3 law. Related phenomenon will be in-between.

Now, let's move on to the real meat of this subject:
Normalcy.

There is a perception amongst so many of us that "nothing ever happens". This is despite the fact that we are living through a nuclear war and what looks like the final chapters of revelations.

How can that be?
Every single day we are on edge with events going around the world, absorbing so much information and reading so many story. Sometimes witnessing man made horrors beyond our comprehension.

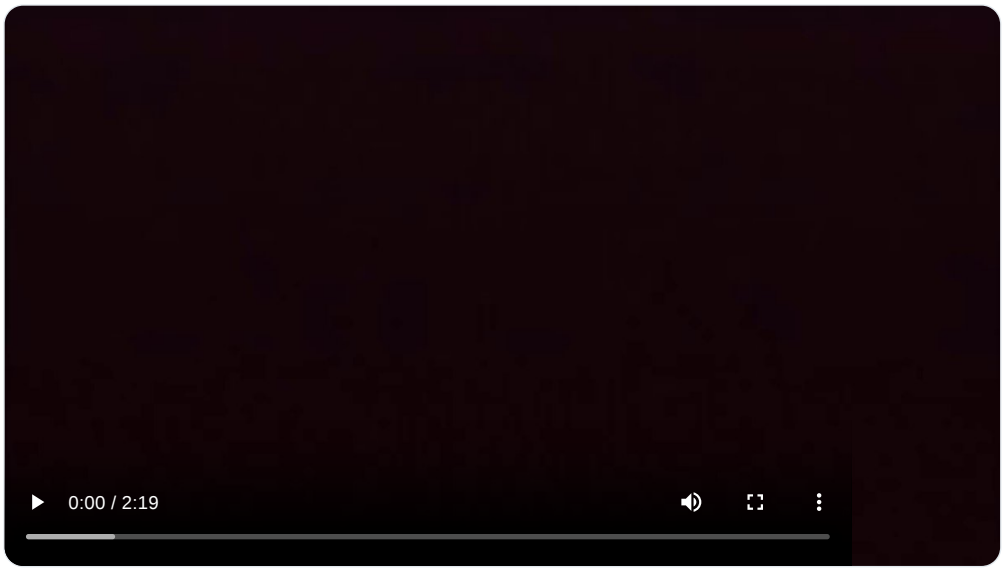
These are all changes that we accept after some difficulty... then they become normal.

All further experiences are compared to these changes. After all, is not time merely a measure of change? And did we not establish that we perceive time in a logarithmic fashion?

What about escalation and our decision making?
Moral depravity? And so on?

We humans need to be careful as our perception is quite skewed towards normalising our experiences.
The more intense the satanic light in front of us, the more desensitised we become towards it.
The more we escalate the more our appetite grows for more escalation.

Some of us, with perfect recollection of the disasters of the past, can mentally stop time though, and net out some advice that others should listen to.
Isn't it time you stopped provoking Russia, NATO? Do you need to relive the horrors of WWII with nukes?
See you next time.
/End



• • •